

CA11

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Question 1

1)

Table 1: Row profiles

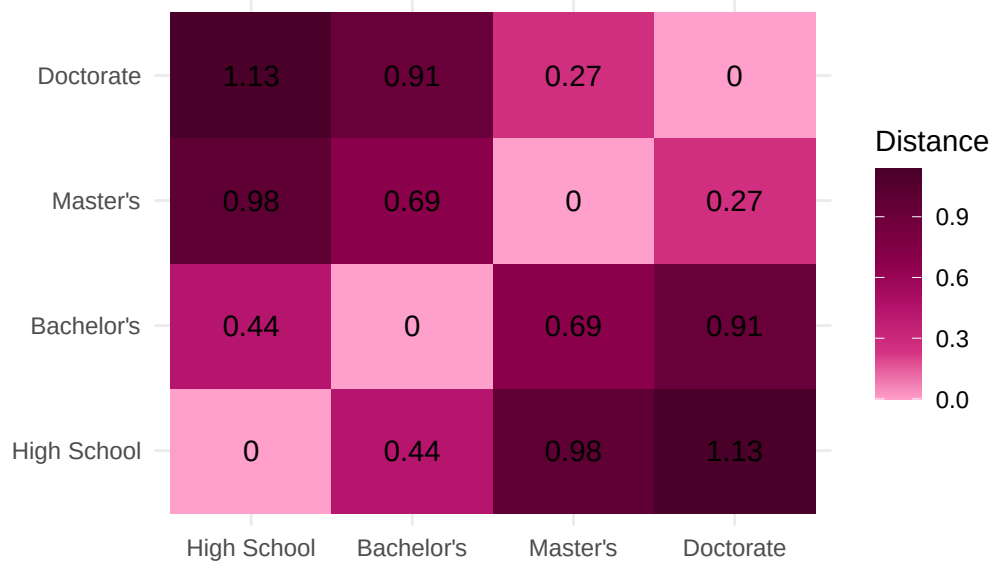
Action	Drama	Comedy	Documentary
0.3125	0.3750	0.2500	0.0625
0.1500	0.4000	0.3000	0.1500
0.0833	0.3333	0.1667	0.4167
0.1250	0.2500	0.1250	0.5000

Table 2: Column profiles

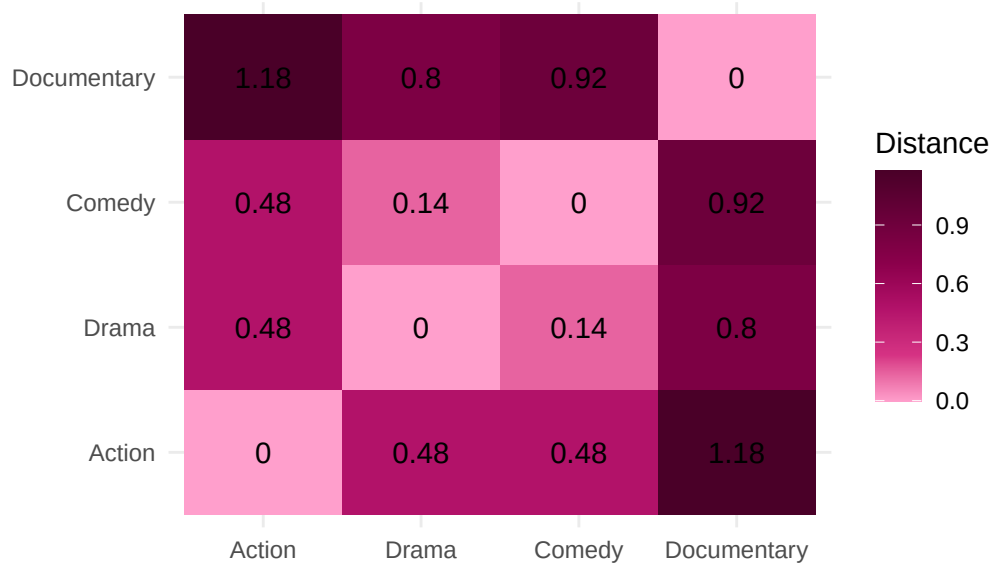
High.School	Bachelor.s	Master.s	Doctorate
0.5495	0.3297	0.1099	0.0110
0.3297	0.4396	0.2198	0.0110
0.3306	0.4959	0.1653	0.0083
0.1064	0.3191	0.5319	0.0426

2)

Row distance heatmap.



Column distance heatmap.



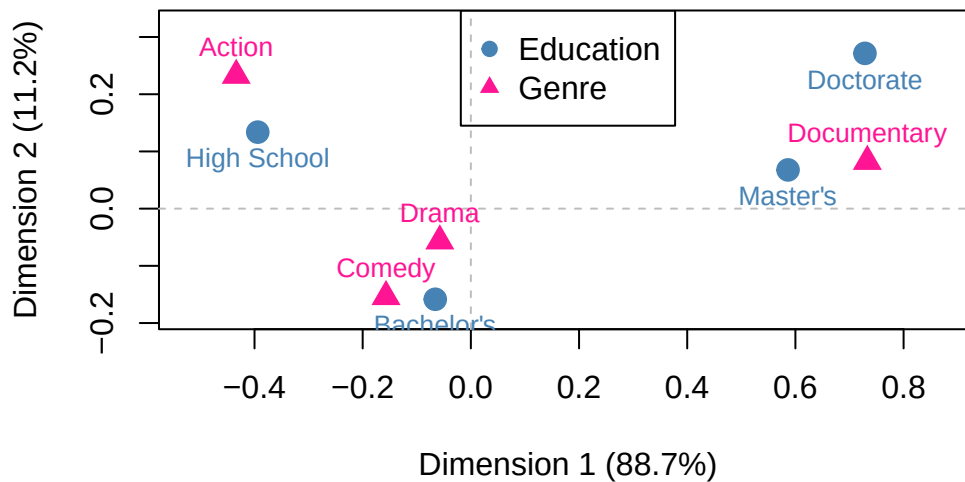
3)

Rows: Doctorate and High school > Master's and High School > Doctorate and Bachelors

Columns: Documentary and Action > Documentary and Comedy > Documentary and Drama

4)

Principal coordinate plot.



Dimension 1 is responsible for 88.7% of the association in this plot, so looking left to right tells us more than up and down as dimension 2 is only responsible for 11.2%. We can clearly see that a lower standard of education results in a closer association with action, comedy and drama. In particular we see that high school is associated with action, bachelors is associated with drama and comedy. Then both Masters and doctorate are associated with documentaries however we note that Masters is close to the other movie categories. We can also see that Doctorate is further away from documentary than Masters is, meaning that masters are more closely associated to documentaries than doctorates (Maybe its because doctorates prefer books.)

5)

Principal inertias (eigenvalues):

dim	value	%	cum%	scree plot
1	0.145916	88.7	88.7	*****
2	0.018494	11.2	99.9	***
3	0.000148	0.1	100.0	

Total: 0.164558 100.0				

Rows:

	name	mass	qlt	inr	k=1 cor	ctr	k=2 cor	ctr
1	HghS	328	1000	345	-394	897	349	134 103 317
2	Bchl	410	1000	73	-66	147	12	-158 853 557
3	Mstr	246	1000	521	586	987	579	68 13 61
4	Dctr	16	987	61	729	867	60	271 120 65

Columns:

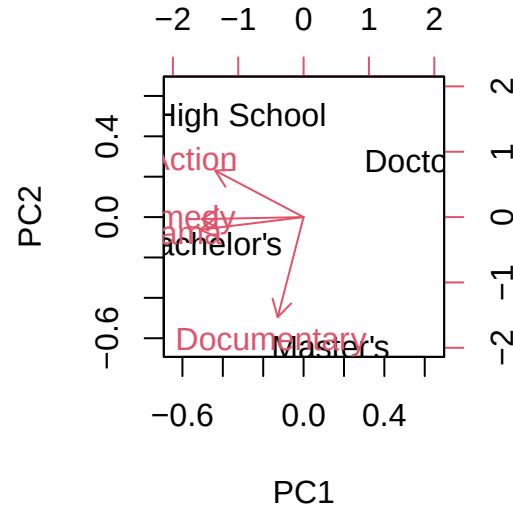
	name	mass	qlt	inr	k=1 cor	ctr	k=2 cor	ctr
1	Actn	186	1000	275	-434	776	241	233 224 547
2	Dram	373	967	15	-57	490	8	-57 478 65
3	Cmdy	248	995	73	-157	507	42	-154 488 318
4	Dcmn	193	1000	637	733	987	709	82 12 71

The dimensions reflect that of the plot with dimension 1 being the most associated and dimension 3 being so small that we can essentially disregard it. As dimensions 1 and 2 account for 99.9% of the association we can say the 2D plot is an accurate representation of the association structure. Mass shows the number of observations as a percentage. We see in rows bachelors is the largest and in columns drama is the largest. Inertia mentions how much that point contributes to the overall association. Masters and Documentary have the highest values for us and thus drive most of our association structure. Cor shows how much quality is explained by that particular dimension for that variable, Ctr tells us how much that particular dimension contributes to the overall inertia of that variable.

Dimension 1 is dominated by Documentary (ctr = 709) on one end and High School/Action on the other, capturing the primary education-genre gradient. Dimension 2 is largely driven by Bachelor's (ctr = 557) and Action (ctr = 547), reflecting a secondary distinction among the lower-education, mainstream-genre cluster. Drama has the lowest inertia, indicating its genre profile is closest to what independence would predict.

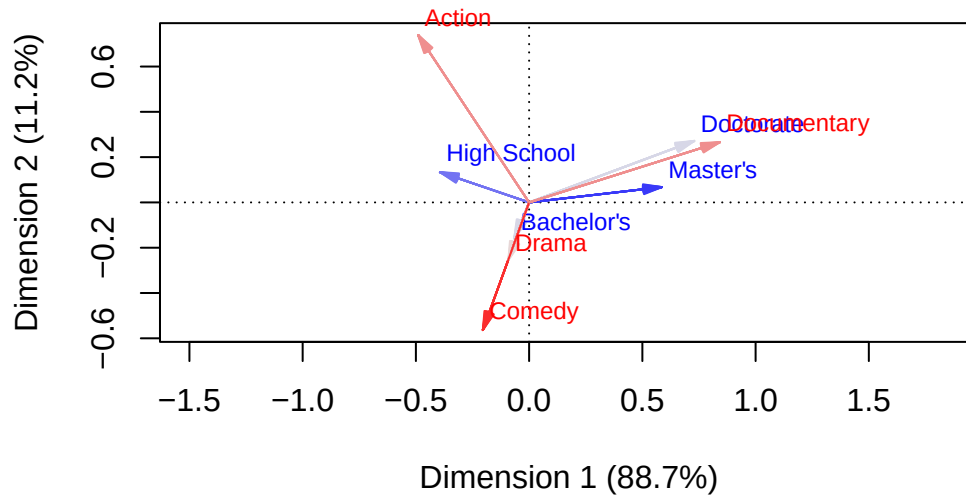
6)

PCA Biplot on Correlation Matrix



The PCA biplot tells a broadly similar story to the CA plot. However, PCA weights all education levels equally regardless of how many observations they represent. Doctorate has only 16 observations compared to Bachelor's 400, yet both have equal leverage on the axes. This inflates Doctorate's influence and pushes it far to the right, whereas CA down-weights it by its low mass, pulling it closer to the centre.

7)



8)

This is the exact reason we use `rowgreen` above instead of `rowprincipal`. When the principal inertias are very unequal, such as in our data where dimension 1 accounts for 88.7% of inertia, the row points get stretched far out along that axis while the column points in standard coordinates remain clustered near the origin, making it difficult to read associations between the two sets of points.

Question 2

a)

Table 3: MCA for HairEyeColor

Hair	Eye	Sex
Black	Brown	Male
Black	Brown	Male
Black	Brown	Male

[illegible]

Hair	Eye	Sex
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Brown	Green	Male
Red	Green	Male
Red	Green	Male
Red	Green	Male
Red	Green	Male
Red	Green	Male
Red	Green	Male
Red	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Blond	Green	Male
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female
Black	Brown	Female

Hair	Eye	Sex
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Brown	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Red	Brown	Female
Blond	Brown	Female
Blond	Brown	Female
Blond	Brown	Female
Blond	Brown	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Black	Blue	Female
Brown	Blue	Female
Brown	Blue	Female
Brown	Blue	Female

[illegible]

Hair	Eye	Sex
Red	Green	Female
Red	Green	Female
Red	Green	Female
Red	Green	Female
Red	Green	Female
Red	Green	Female
Red	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female
Blond	Green	Female

b)

Principal inertias (eigenvalues):

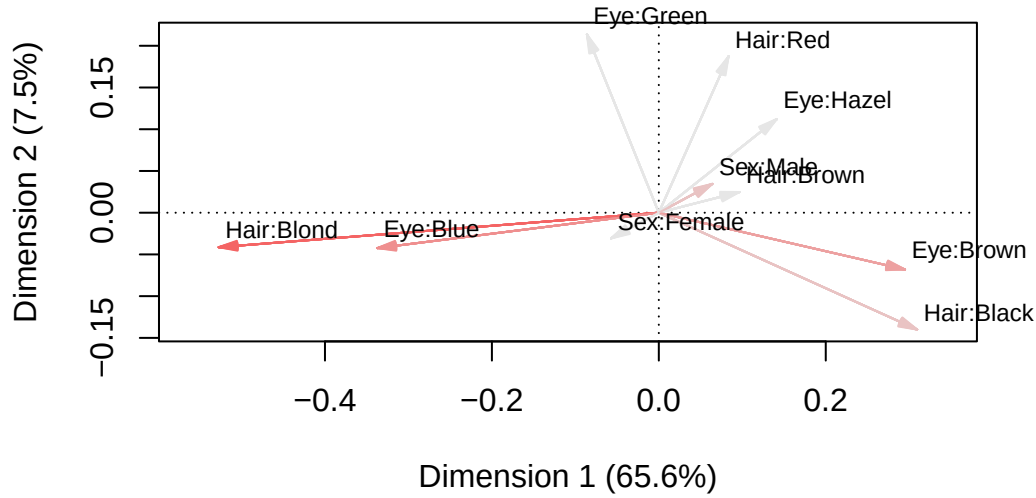
dim	value	%	cum%	scree plot
1	0.054579	65.6	65.6	*****
2	0.006263	7.5	73.1	***
3	0.000871	1.0	74.1	

Total: 0.083229

Columns:

	name	mass	qlt	inr	k=1 cor	ctr	k=2 cor	ctr
1	Hair:Black	61	738	117	310 613	107	-140 125	191
2	Hair:Brown	161	691	71	97 648	27	25 43	16
3	Hair:Red	40	667	119	84 111	5	187 556	224
4	Hair:Blond	72	745	126	-527 741	364	-41 5	19
5	Eye:Brown	124	724	96	295 688	198	-68 37	92
6	Eye:Blue	121	754	100	-337 742	252	-42 12	35
7	Eye:Hazel	52	739	114	141 453	19	112 286	105
8	Eye:Green	36	677	121	-86 95	5	214 582	262
9	Sex:Male	157	588	72	64 456	12	35 132	30

10 | Sex:Female | 176 588 64 | -57 456 11 | -31 132 27 |



Using the lambda-adjusted method, dimension 1 accounts for 65.6% of the association and dimension 2 for 7.5%, giving 73.1% cumulatively. Dimension 1 captures a light-to-dark colouring gradient: Hair:Blond and Eye:Blue are strongly associated on the left, while Hair:Black and Eye:Brown are strongly associated on the right. Dimension 2 separates rarer feature combinations (Hair:Red, Eye:Green) from more common ones. Sex lies close to the origin in both dimensions, indicating it contributes little to the association structure between hair and eye colour. Sex:Female points weakly toward Hair:Blond and Eye:Blue, while Sex:Male points weakly toward Hair:Brown and Eye:Hazel, suggesting a slight tendency for females to have lighter colouring and males darker. However, as both sex categories lie close to the origin, this association is weak and should be interpreted with caution.